MPG No. 1.1.1 1 [Type I, triclinic] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} Q_x & Q_y & Q_z \end{bmatrix}$$

$$Q_x = \mathbb{Q}_x^{(1)}$$
$$Q_y = \mathbb{Q}_y^{(1)}$$
$$Q_z = \mathbb{Q}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{xy} & Q_{yy} & Q_{yz} \\ Q_{xz} & Q_{yz} & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$Q_{xy} = \mathbb{Q}_{xy}^{(1)}$$
$$Q_{xz} = \mathbb{Q}_{xz}^{(1)}$$
$$Q_{yy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$
$$Q_{yz} = \mathbb{Q}_{yz}^{(1)}$$
$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & Q_{xy} & Q_{xz} \\ -Q_{xy} & 0 & Q_{yz} \\ -Q_{xz} & -Q_{yz} & 0 \end{bmatrix}$$

$$Q_{xy} = \mathbb{G}_z^{(1)}$$
$$Q_{xz} = -\mathbb{G}_y^{(1)}$$
$$Q_{yz} = \mathbb{G}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} Q_{xxx} & Q_{xxy} & Q_{xxz} \\ Q_{yyx} & Q_{yyy} & Q_{yyz} \\ Q_{zzx} & Q_{zzy} & Q_{zzz} \\ Q_{yzx} & Q_{yyz} & Q_{zzy} \\ Q_{xxz} & Q_{yzx} & Q_{zzx} \\ Q_{xxy} & Q_{yyx} & Q_{yzx} \end{bmatrix}$$

$$
\begin{aligned}
Q_{xxx} &= 3\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)} \\
Q_{xxy} &= \mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{xxz} &= \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\beta(1)} \\
Q_{yyx} &= \mathbb{Q}_x^{(1)} - \mathbb{Q}_x^{\alpha(1)} + \mathbb{Q}_x^{\beta(1)} \\
Q_{yyy} &= 3\mathbb{Q}_y^{(1)} + 2\mathbb{Q}_y^{\alpha(1)} \\
Q_{yyz} &= \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\beta(1)} \\
Q_{zzx} &= \mathbb{Q}_x^{(1)} - \mathbb{Q}_x^{\alpha(1)} - \mathbb{Q}_x^{\beta(1)} \\
Q_{zzy} &= \mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\beta(1)} \\
Q_{zzz} &= 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} \\
Q_{yzx} &= \mathbb{Q}_{xyz}^{(1)}
\end{aligned}
$$

* Rank 3 tensor (sa).

$$
\begin{bmatrix}
0 & Q_{xxy} & Q_{xxz} \\
Q_{yyx} & 0 & Q_{yyz} \\
Q_{zzx} & Q_{zzy} & 0 \\
Q_{yzx} & -\frac{Q_{yyz}}{2} & -\frac{Q_{zzy}}{2} \\
-\frac{Q_{xxz}}{2} & Q_{zxy} & -\frac{Q_{zzx}}{2} \\
-\frac{Q_{xxy}}{2} & -\frac{Q_{yyx}}{2} & -Q_{yzx} - Q_{zxy}
\end{bmatrix}
$$

$$
\begin{aligned}
Q_{xxy} &= -2\mathbb{G}_{xz}^{(1)} + 2\mathbb{Q}_y^{(2)} \\
Q_{xxz} &= 2\mathbb{G}_{xy}^{(1)} + 2\mathbb{Q}_z^{(2)} \\
Q_{yyx} &= 2\mathbb{G}_{yz}^{(1)} + 2\mathbb{Q}_x^{(2)} \\
Q_{yyz} &= -2\mathbb{G}_{xy}^{(1)} + 2\mathbb{Q}_z^{(2)} \\
Q_{zzx} &= -2\mathbb{G}_{yz}^{(1)} + 2\mathbb{Q}_x^{(2)} \\
Q_{zzy} &= 2\mathbb{G}_{xz}^{(1)} + 2\mathbb{Q}_y^{(2)} \\
Q_{yzx} &= 3\mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)} \\
\frac{Q_{yyz}}{2} &= -\mathbb{G}_{xy}^{(1)} + \mathbb{Q}_z^{(2)} \\
\frac{Q_{zzy}}{2} &= \mathbb{G}_{xz}^{(1)} + \mathbb{Q}_y^{(2)} \\
\frac{Q_{xxz}}{2} &= \mathbb{G}_{xy}^{(1)} + \mathbb{Q}_z^{(2)} \\
Q_{zxy} &= -3\mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)} \\
\frac{Q_{zzx}}{2} &= -\mathbb{G}_{yz}^{(1)} + \mathbb{Q}_x^{(2)} \\
\frac{Q_{xxy}}{2} &= -\mathbb{G}_{xz}^{(1)} + \mathbb{Q}_y^{(2)} \\
\frac{Q_{yyx}}{2} &= \mathbb{G}_{yz}^{(1)} + \mathbb{Q}_x^{(2)}
\end{aligned}
$$

* Rank 3 tensor (a).

$$
\begin{bmatrix}
Q_{yzx} & Q_{yzy} & Q_{yzz} \\
Q_{zxx} & Q_{zxy} & Q_{zxz} \\
Q_{xyx} & Q_{xyy} & Q_{xyz}
\end{bmatrix}
$$

$$Q_{yzx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(2)} + \mathbb{G}_v^{(2)}$$
$$Q_{yzy} = \mathbb{G}_{xy}^{(2)} + \mathbb{Q}_z^{(3)}$$
$$Q_{yzz} = \mathbb{G}_{xz}^{(2)} - \mathbb{Q}_y^{(3)}$$
$$Q_{zxx} = \mathbb{G}_{xy}^{(2)} - \mathbb{Q}_z^{(3)}$$
$$Q_{zxy} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(2)} - \mathbb{G}_v^{(2)}$$
$$Q_{zxz} = \mathbb{G}_{yz}^{(2)} + \mathbb{Q}_x^{(3)}$$
$$Q_{xyx} = \mathbb{G}_{xz}^{(2)} + \mathbb{Q}_y^{(3)}$$
$$Q_{xyy} = \mathbb{G}_{yz}^{(2)} - \mathbb{Q}_x^{(3)}$$
$$Q_{xyz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & Q_{yyyz} & Q_{yyzx} & Q_{yyxy} \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & Q_{zzyz} & Q_{zzzx} & Q_{zzxy} \\ Q_{xxyz} & Q_{yyyz} & Q_{zzyz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{xxzx} & Q_{yyzx} & Q_{zzzx} & Q_{yzzx} & Q_{zzzx} & Q_{zxxy} \\ Q_{xxxy} & Q_{yyxy} & Q_{zzxy} & Q_{yzxy} & Q_{zxxy} & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)}$$
$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$Q_{xxyz} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)}$$
$$Q_{xxzx} = -\mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)}$$
$$Q_{xxxy} = \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)}$$
$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$
$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$
$$Q_{yyyz} = \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)}$$
$$Q_{yyzx} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)}$$
$$Q_{yyxy} = -\mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)}$$
$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{zzyz} = -\mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)}$$
$$Q_{zzzx} = \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)}$$
$$Q_{zzxy} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$
$$Q_{yzzx} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)}$$
$$Q_{yzxy} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)}$$
$$Q_{zxzx} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$
$$Q_{zxxy} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(2)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ -Q_{xxyy} & 0 & Q_{yyzz} & Q_{yyyz} & Q_{yyzx} & Q_{yyxy} \\ -Q_{xxzz} & -Q_{yyzz} & 0 & Q_{zzyz} & Q_{zzzx} & Q_{zzxy} \\ -Q_{xxyz} & -Q_{yyyz} & -Q_{zzyz} & 0 & Q_{yzzx} & Q_{yzxy} \\ -Q_{xxzx} & -Q_{yyzx} & -Q_{zzzx} & -Q_{yzzx} & 0 & Q_{zxxy} \\ -Q_{xxxy} & -Q_{yyxy} & -Q_{zzxy} & -Q_{yzxy} & -Q_{zxxy} & 0 \end{bmatrix}$$

$$Q_{xxyy} = 4\mathbb{G}^{(1)}_{xyz} - 2\mathbb{Q}^{(3)}_{v}$$
$$Q_{xxzz} = -4\mathbb{G}^{(1)}_{xyz} + 3\mathbb{Q}^{(3)}_{u} - \mathbb{Q}^{(3)}_{v}$$
$$Q_{xxyz} = -4\mathbb{G}^{\beta(1)}_{x} + \mathbb{Q}^{(3)}_{yz}$$
$$Q_{xxzx} = -2\mathbb{G}^{(1)}_{y} + 2\mathbb{G}^{\alpha(1)}_{y} + 2\mathbb{G}^{\beta(1)}_{y} + \mathbb{Q}^{(3)}_{xz}$$
$$Q_{xxxy} = 2\mathbb{G}^{(1)}_{z} - 2\mathbb{G}^{\alpha(1)}_{z} + 2\mathbb{G}^{\beta(1)}_{z} + \mathbb{Q}^{(3)}_{xy}$$
$$Q_{yyzz} = 4\mathbb{G}^{(1)}_{xyz} + 3\mathbb{Q}^{(3)}_{u} + \mathbb{Q}^{(3)}_{v}$$
$$Q_{yyyz} = 2\mathbb{G}^{(1)}_{x} - 2\mathbb{G}^{\alpha(1)}_{x} + 2\mathbb{G}^{\beta(1)}_{x} + \mathbb{Q}^{(3)}_{yz}$$
$$Q_{yyzx} = -4\mathbb{G}^{\beta(1)}_{y} + \mathbb{Q}^{(3)}_{xz}$$
$$Q_{yyxy} = -2\mathbb{G}^{(1)}_{z} + 2\mathbb{G}^{\alpha(1)}_{z} + 2\mathbb{G}^{\beta(1)}_{z} + \mathbb{Q}^{(3)}_{xy}$$
$$Q_{zzyz} = -2\mathbb{G}^{(1)}_{x} + 2\mathbb{G}^{\alpha(1)}_{x} + 2\mathbb{G}^{\beta(1)}_{x} + \mathbb{Q}^{(3)}_{yz}$$
$$Q_{zzzx} = 2\mathbb{G}^{(1)}_{y} - 2\mathbb{G}^{\alpha(1)}_{y} + 2\mathbb{G}^{\beta(1)}_{y} + \mathbb{Q}^{(3)}_{xz}$$
$$Q_{zzxy} = -4\mathbb{G}^{\beta(1)}_{z} + \mathbb{Q}^{(3)}_{xy}$$
$$Q_{yzzx} = -\mathbb{G}^{(1)}_{z} - 4\mathbb{G}^{\alpha(1)}_{z}$$
$$Q_{yzxy} = \mathbb{G}^{(1)}_{y} + 4\mathbb{G}^{\alpha(1)}_{y}$$
$$Q_{zxxy} = -\mathbb{G}^{(1)}_{x} - 4\mathbb{G}^{\alpha(1)}_{x}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{yzzx} & Q_{zxzx} & Q_{zxxy} \\ Q_{yzxy} & Q_{zxxy} & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(4)}_{v}$$
$$Q_{yzzx} = 2\mathbb{Q}^{(4)}_{xy}$$
$$Q_{yzxy} = 2\mathbb{Q}^{(4)}_{xz}$$
$$Q_{zxzx} = \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} - 2\mathbb{Q}^{(4)}_{v}$$
$$Q_{zxxy} = 2\mathbb{Q}^{(4)}_{yz}$$
$$Q_{xyxy} = \mathbb{Q}^{(3)}_{0} + 4\mathbb{Q}^{(4)}_{u}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & Q_{yzzx} & Q_{yzxy} \\ -Q_{yzzx} & 0 & Q_{zxxy} \\ -Q_{yzxy} & -Q_{zxxy} & 0 \end{bmatrix}$$

$$Q_{yzzx} = -\mathbb{G}^{(2)}_{z}$$
$$Q_{yzxy} = \mathbb{G}^{(2)}_{y}$$
$$Q_{zxxy} = -\mathbb{G}^{(2)}_{x}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ Q_{yyyz} & Q_{yyzx} & Q_{yyxy} \\ Q_{zzyz} & Q_{zzzx} & Q_{zzxy} \\ Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{zxyz} & Q_{zxzx} & Q_{zxxy} \\ Q_{xyyz} & Q_{xyzx} & Q_{xyxy} \end{bmatrix}$$

$$
\begin{aligned}
Q_{xxyz} &= \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{(4)} + 2\mathbb{G}_x^{\alpha(2)} \\
Q_{xxzx} &= \mathbb{G}_y^{(3)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{xz}^{(5)} \\
Q_{xxxy} &= \mathbb{G}_z^{(3)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} - 2\mathbb{Q}_{xy}^{(5)} \\
Q_{yyyz} &= \mathbb{G}_x^{(3)} - \mathbb{G}_x^{\alpha(2)} + \mathbb{G}_x^{\beta(2)} - 2\mathbb{Q}_{yz}^{(5)} \\
Q_{yyzx} &= \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)} \\
Q_{yyxy} &= \mathbb{G}_z^{(3)} - \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{xy}^{(5)} \\
Q_{zzyz} &= \mathbb{G}_x^{(3)} - \mathbb{G}_x^{\alpha(2)} - \mathbb{G}_x^{\beta(2)} + 2\mathbb{Q}_{yz}^{(5)} \\
Q_{zzzx} &= \mathbb{G}_y^{(3)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} - 2\mathbb{Q}_{xz}^{(5)} \\
Q_{zzxy} &= \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)} \\
Q_{yzyz} &= \mathbb{G}_{xyz}^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)} \\
Q_{yzzx} &= \mathbb{G}_z^{(4)} - \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} - \mathbb{Q}_{xy}^{(5)} \\
Q_{yzxy} &= \mathbb{G}_y^{(4)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(5)} \\
Q_{zxyz} &= \mathbb{G}_z^{(4)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} + \mathbb{Q}_{xy}^{(5)} \\
Q_{zxzx} &= \mathbb{G}_{xyz}^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)} \\
Q_{zxxy} &= \mathbb{G}_x^{(4)} - \mathbb{G}_x^{\alpha(2)} - \mathbb{G}_x^{\beta(2)} - \mathbb{Q}_{yz}^{(5)} \\
Q_{xyyz} &= \mathbb{G}_y^{(4)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{xz}^{(5)} \\
Q_{xyzx} &= \mathbb{G}_x^{(4)} - \mathbb{G}_x^{\alpha(2)} + \mathbb{G}_x^{\beta(2)} + \mathbb{Q}_{yz}^{(5)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(2)} + 2\mathbb{Q}_v^{(5)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
Q_{yzxx} & Q_{yzyy} & Q_{yzzz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\
Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & Q_{zxyz} & Q_{zxzx} & Q_{zxxy} \\
Q_{xyxx} & Q_{xyyy} & Q_{xyzz} & Q_{xyyz} & Q_{xyzx} & Q_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
Q_{yzxx} &= \mathbb{G}_x^{(5)} + 2\mathbb{G}_x^{(6)} + 2\mathbb{G}_x^{\alpha(3)} \\
Q_{yzyy} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzzz} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzyz} &= \mathbb{G}_{xyz}^{(3)} - 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)} \\
Q_{yzzx} &= \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} + \mathbb{Q}_{xy}^{(6)} \\
Q_{yzxy} &= \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{zxxx} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyy} &= \mathbb{G}_y^{(5)} + 2\mathbb{G}_y^{(6)} + 2\mathbb{G}_y^{\alpha(3)} \\
Q_{zxzz} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyz} &= \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} - \mathbb{Q}_{xy}^{(6)} \\
Q_{zxzx} &= \mathbb{G}_{xyz}^{(3)} + 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)} \\
Q_{zxxy} &= \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} + \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxx} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyyy} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyzz} &= \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)} \\
Q_{xyyz} &= \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyzx} &= \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} - \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(3)} + 2\mathbb{Q}_v^{(6)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} & Q_{xxzy} & Q_{xxxz} & Q_{xxyx} \\
Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & Q_{yyyz} & Q_{yyzx} & Q_{yyxy} & Q_{yyzy} & Q_{yyxz} & Q_{yyyx} \\
Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & Q_{zzyz} & Q_{zzzx} & Q_{zzxy} & Q_{zzzy} & Q_{zzxz} & Q_{zzyx} \\
Q_{yzxx} & Q_{yzyy} & Q_{yzzz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzy} & Q_{yzxz} & Q_{yzyx} \\
Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & Q_{zxyz} & Q_{zxzx} & Q_{zxxy} & Q_{zxzy} & Q_{zxxz} & Q_{zxyx} \\
Q_{xyxx} & Q_{xyyy} & Q_{xyzz} & Q_{xyyz} & Q_{xyzx} & Q_{xyxy} & Q_{xyzy} & Q_{xyxz} & Q_{xyyx}
\end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$

$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$

$$Q_{xxzz} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$

$$Q_{xxyz} = \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{(4)} + 2\mathbb{G}_x^{\alpha(2)} - 4\mathbb{G}_x^{\beta(1)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + \mathbb{Q}_{yz}^{(3)}$$

$$Q_{xxzx} = -2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{xxxy} = 2\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{G}_z^{\beta(2)} + \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{xxzy} = -\mathbb{G}_x^{(3)} - 2\mathbb{G}_x^{(4)} - 2\mathbb{G}_x^{\alpha(2)} - 4\mathbb{G}_x^{\beta(1)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + \mathbb{Q}_{yz}^{(3)}$$

$$Q_{xxxz} = -2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{xxyx} = 2\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{G}_z^{\beta(2)} + \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} + 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{yyxx} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} + 2\mathbb{Q}_v^{(3)}$$

$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$

$$Q_{yyzz} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$

$$Q_{yyyz} = 2\mathbb{G}_x^{(1)} + \mathbb{G}_x^{(3)} - 2\mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_x^{\beta(2)} + \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{yyzx} = \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$

$$Q_{yyxy} = -2\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{G}_z^{\beta(2)} - \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} + 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{yyzy} = 2\mathbb{G}_x^{(1)} - \mathbb{G}_x^{(3)} - 2\mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} - \mathbb{G}_x^{\beta(2)} + \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} + 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{yyxz} = -\mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{(4)} - 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$

$$Q_{yyyx} = -2\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{G}_z^{\beta(2)} - \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{zzxx} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$

$$Q_{zzyy} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{zzyz} = -2\mathbb{G}_x^{(1)} + \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} - \mathbb{G}_x^{\beta(2)} - \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} + 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{zzzx} = 2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)} - 4\mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + \mathbb{Q}_{xy}^{(3)}$$

$$Q_{zzzy} = -2\mathbb{G}_x^{(1)} - \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_x^{\beta(2)} - \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{zzxz} = 2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{zzyx} = -\mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{(4)} - 2\mathbb{G}_z^{\alpha(2)} - 4\mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + \mathbb{Q}_{xy}^{(3)}$$

$$Q_{yzxx} = 4\mathbb{G}_x^{\beta(1)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} - \mathbb{Q}_{yz}^{(3)}$$

$$Q_{yzyy} = -2\mathbb{G}_x^{(1)} + 2\mathbb{G}_x^{\alpha(1)} - 2\mathbb{G}_x^{\beta(1)} + \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(3)}$$

$$Q_{yzzz} = 2\mathbb{G}_x^{(1)} - 2\mathbb{G}_x^{\alpha(1)} - 2\mathbb{G}_x^{\beta(1)} - \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(3)}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(5)}$$

$$Q_{yzxy} = \mathbb{G}_y^{(1)} + \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(5)}$$

$$Q_{yzzy} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$

$$Q_{yzxz} = -\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(5)}$$

$$Q_{yzyx} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)}$$

$$Q_{zxxx} = 2\mathbb{G}_y^{(1)} - 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$

$$Q_{zxyy} = 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} - \mathbb{Q}_{xz}^{(3)}$$

$$Q_{zxzz} = -2\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$

$$Q_{zxyz} = \mathbb{G}_z^{(1)} + \mathbb{G}_z^{(4)} + 4\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(5)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$

$$Q_{zxxy} = -\mathbb{G}_x^{(1)} + \mathbb{G}_x^{(4)} - 4\mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\alpha(2)} - \mathbb{G}_x^{\beta(2)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(5)}$$

$$Q_{zxzy} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{(4)} + 4\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(5)}$$

$$Q_{zxxz} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} Q_{yzxx} & Q_{yzyy} & Q_{yzzz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzy} & Q_{yzxz} & Q_{yzyx} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & Q_{zxyz} & Q_{zxzx} & Q_{zxxy} & Q_{zxzy} & Q_{zxxz} & Q_{zxyx} \\ Q_{xyxx} & Q_{xyyy} & Q_{xyzz} & Q_{xyyz} & Q_{xyzx} & Q_{xyxy} & Q_{xyzy} & Q_{xyxz} & Q_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzxx} &= \mathbb{G}_x^{(5)} + 2\mathbb{G}_x^{(6)} + 2\mathbb{G}_x^{\alpha(3)} \\
Q_{yzyy} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzzz} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzyz} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{yzzx} &= -\mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(4)} + \mathbb{Q}_{xy}^{(6)} \\
Q_{yzxy} &= \mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(4)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{yzzy} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{yzxz} &= \mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(4)} + \mathbb{Q}_{xy}^{(6)} \\
Q_{yzyx} &= -\mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(4)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{zxxx} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyy} &= \mathbb{G}_y^{(5)} + 2\mathbb{G}_y^{(6)} + 2\mathbb{G}_y^{\alpha(3)} \\
Q_{zxzz} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyz} &= \mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(4)} - \mathbb{Q}_{xy}^{(6)} \\
Q_{zxzx} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{zxxy} &= -\mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(4)} + \mathbb{Q}_{yz}^{(6)} \\
Q_{zxzy} &= -\mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(4)} - \mathbb{Q}_{xy}^{(6)} \\
Q_{zxxz} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{zxyx} &= \mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(4)} + \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxx} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyyy} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyzz} &= \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)} \\
Q_{xyyz} &= -\mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(4)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyzx} &= \mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(4)} - \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)} \\
Q_{xyzy} &= \mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(4)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyxz} &= -\mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(4)} - \mathbb{Q}_{yz}^{(6)} \\
Q_{xyyx} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} - 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxxy} & Q_{xxxz} \\ Q_{yyyx} & Q_{yyyy} & Q_{yyyz} \\ Q_{zzzx} & Q_{zzzy} & Q_{zzzz} \\ Q_{yyzx} & Q_{yyzy} & Q_{yyzz} \\ Q_{zzxx} & Q_{zzxy} & Q_{zzxz} \\ Q_{xxyx} & Q_{xxyy} & Q_{xxyz} \\ Q_{yzzx} & Q_{yzzy} & Q_{yzzz} \\ Q_{zxxx} & Q_{zxxy} & Q_{zxxz} \\ Q_{xxyy} + Q_{yyzz} - Q_{yzzy} - Q_{zxxz} + Q_{zzxx} & Q_{xyyy} & Q_{xyyz} \\ -\frac{Q_{yyyz}}{2} + \frac{Q_{yyzy}}{2} + Q_{zxxy} & Q_{xyyz} + \frac{Q_{zzzz}}{2} - \frac{Q_{zzzx}}{2} & \frac{Q_{xxxy}}{2} - \frac{Q_{xxyx}}{2} + Q_{xyyy} - Q_{yyyx} + Q_{zzxy} \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$
$$Q_{xxxy} = \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{xxxz} = -\mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{yyyx} = -\mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{yyyy} = 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$
$$Q_{yyyz} = \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{zzzx} = \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{zzzy} = -\mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{zzzz} = 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{yyzx} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{yyzy} = \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} + 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{zzxy} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + \mathbb{Q}_{xy}^{(3)}$$
$$Q_{zzxz} = \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{xxyx} = \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} + 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$
$$Q_{xxyz} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + \mathbb{Q}_{yz}^{(3)}$$
$$Q_{yzzx} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(5)}$$
$$Q_{yzzy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{yzzz} = -\mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(3)}$$
$$Q_{zxxx} = -\mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zxxy} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(5)}$$
$$Q_{zxxz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{xyyy} = -\mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(3)}$$
$$Q_{xyyz} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)}$$